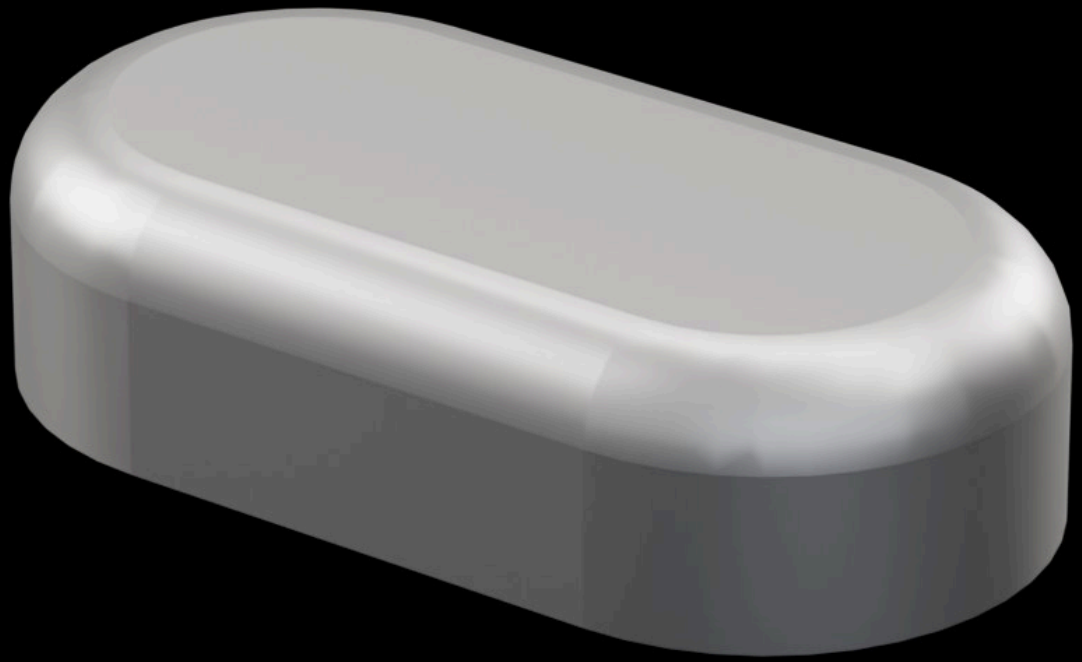
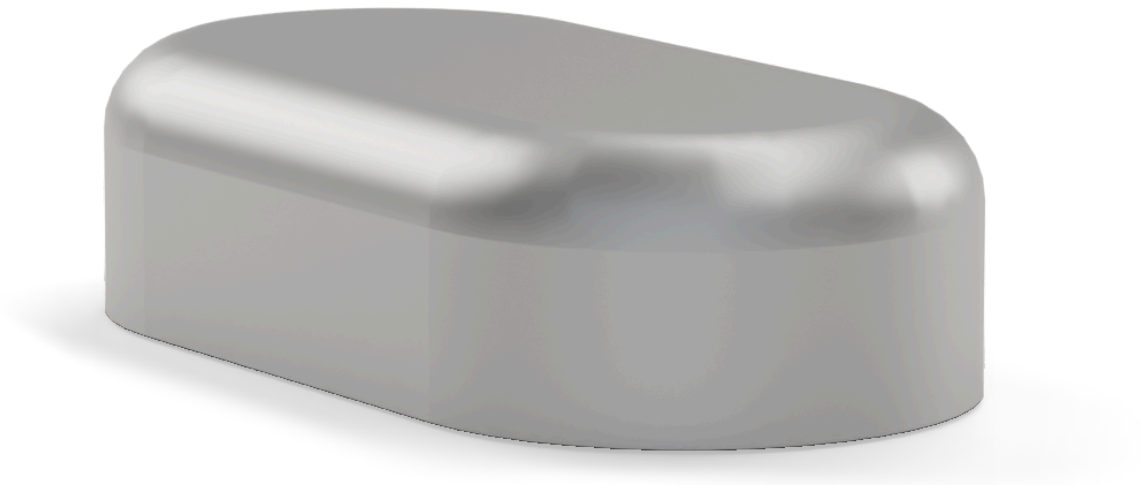


Atrium

Read the body with light.

By Local Forge · Product brief · November 2026 · Internal

Twenty-one millimetres.
Pulse, sleep and skin temperature.



20.9 × 11 × 5.3 mm
One continuous capsule

About 5 days
Per charge, simulated

No account. No cloud.
Readings stay on your phone

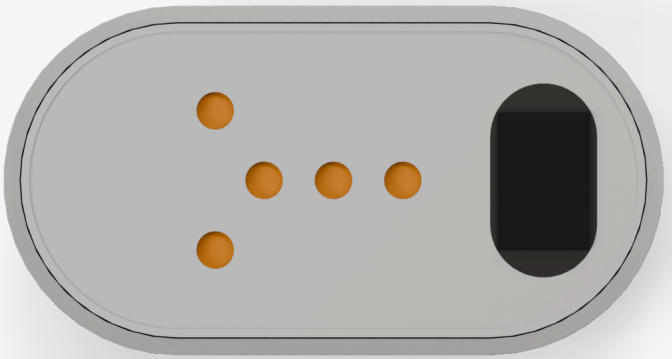
Design

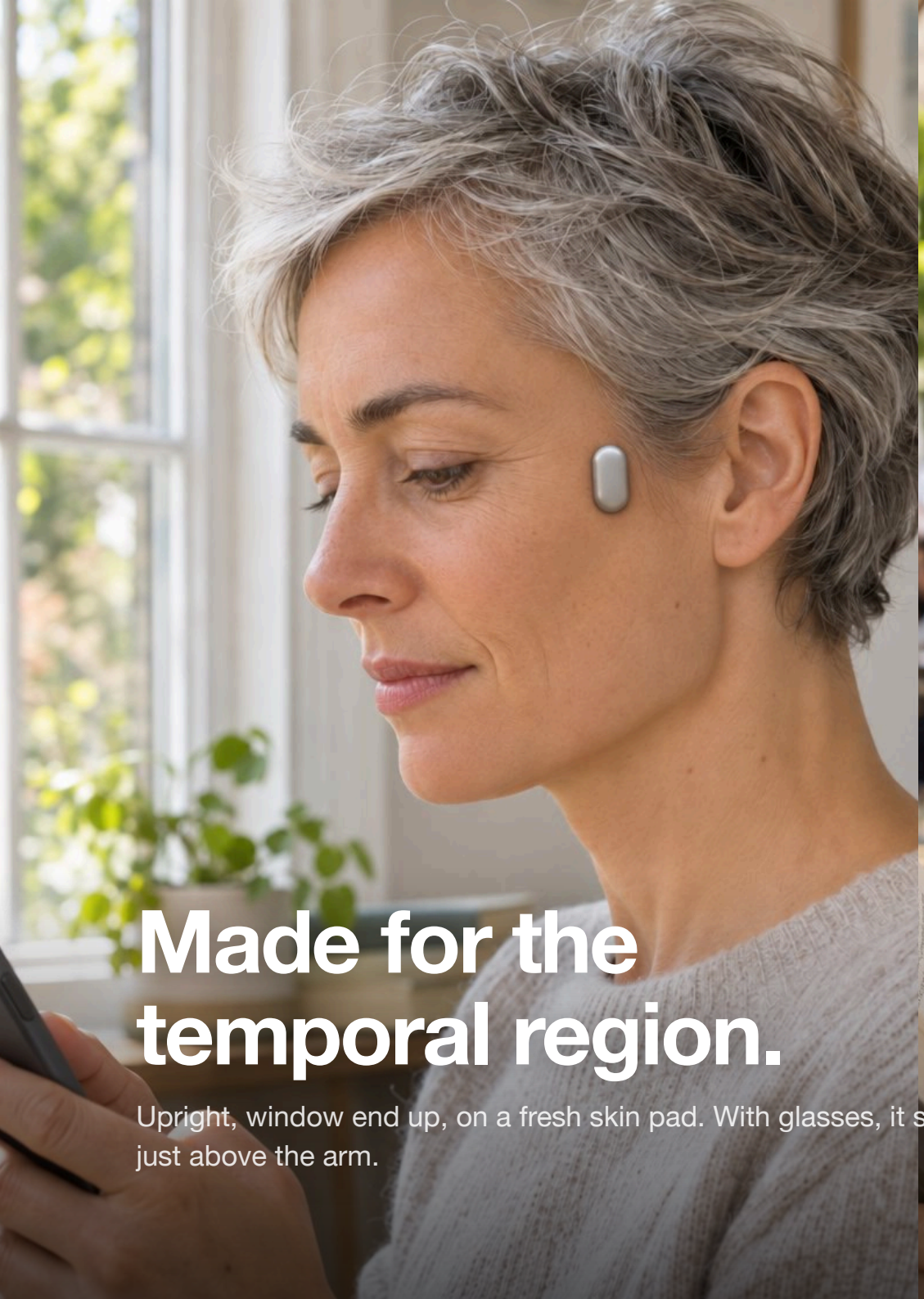
Satin silver. Clear to the antenna.

A moulded polycarbonate shell, finished in non-conductive vacuum metallisation under a matte hardcoat. It looks and feels like bead-blasted aluminium, yet the Bluetooth signal passes straight through to the chip antenna 0.3 mm inside.

No buttons. No lights. No seams. It wakes on skin and pairs when lifted from its Plinth.









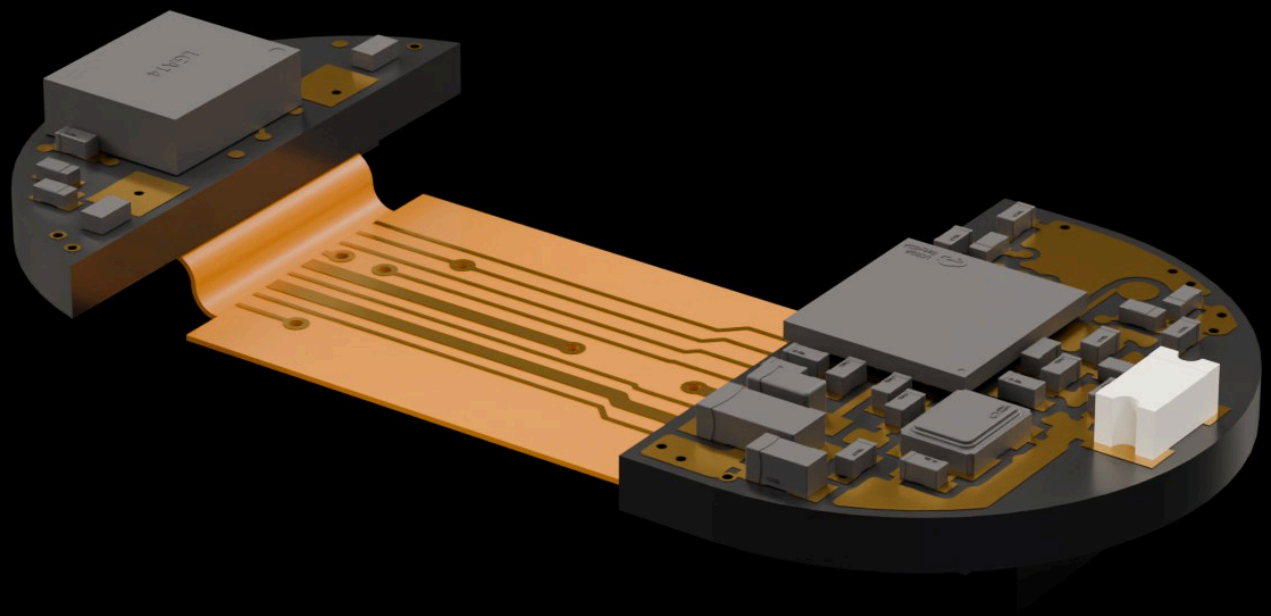
Made for the temporal region.

Upright, window end up, on a fresh skin pad. With glasses, it sits just above the arm.

Inside

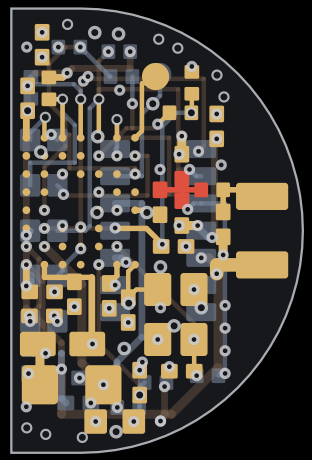
Two islands. One flex. One cell.

Six-layer HDI boards in each rounded end, joined by a flex. In the pod it folds beneath the 30 mAh cell, and the stack is 5.3 mm.

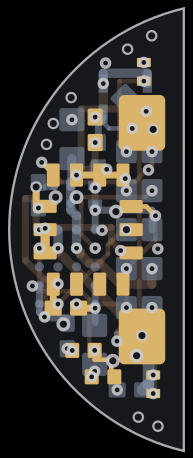


Circuit boards

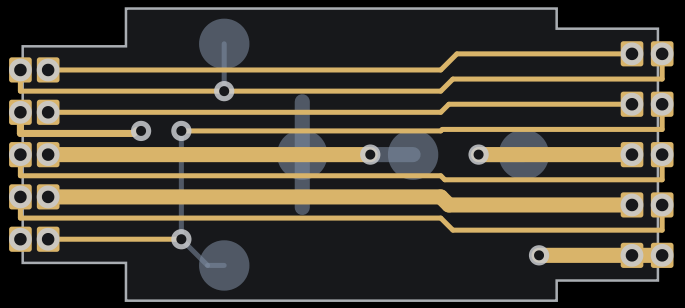
Every trace placed by hand.



Radio island
nRF52832, optics, antenna



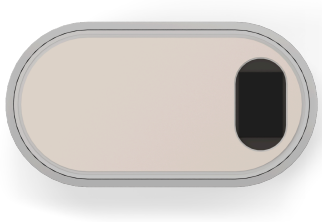
Power island
Charger, protection, motion



Flex jumper
Folds under the cell

6 layers 0.82 mm 318 tracks 186 vias 0 design-rule errors

Everything it needs.
Nothing it doesn't.



Skin pad

Soft, with a window for the optics.
About two days each.



Plinth

Charges it either way round.



Charging case

About eight full charges.



In the box

Pod, Plinth and a card of pads.





Charging case

A pocket charger. And a lifeline.

A 500 mAh cell refills the pod about eight times. The case can also reach the pod's debug port through its contacts, so it can reset or reflash a pod that stops responding.

36 × 25 × 17 mm
Case

2.5 h
To charge the case

USB-C
One button, one light

The app

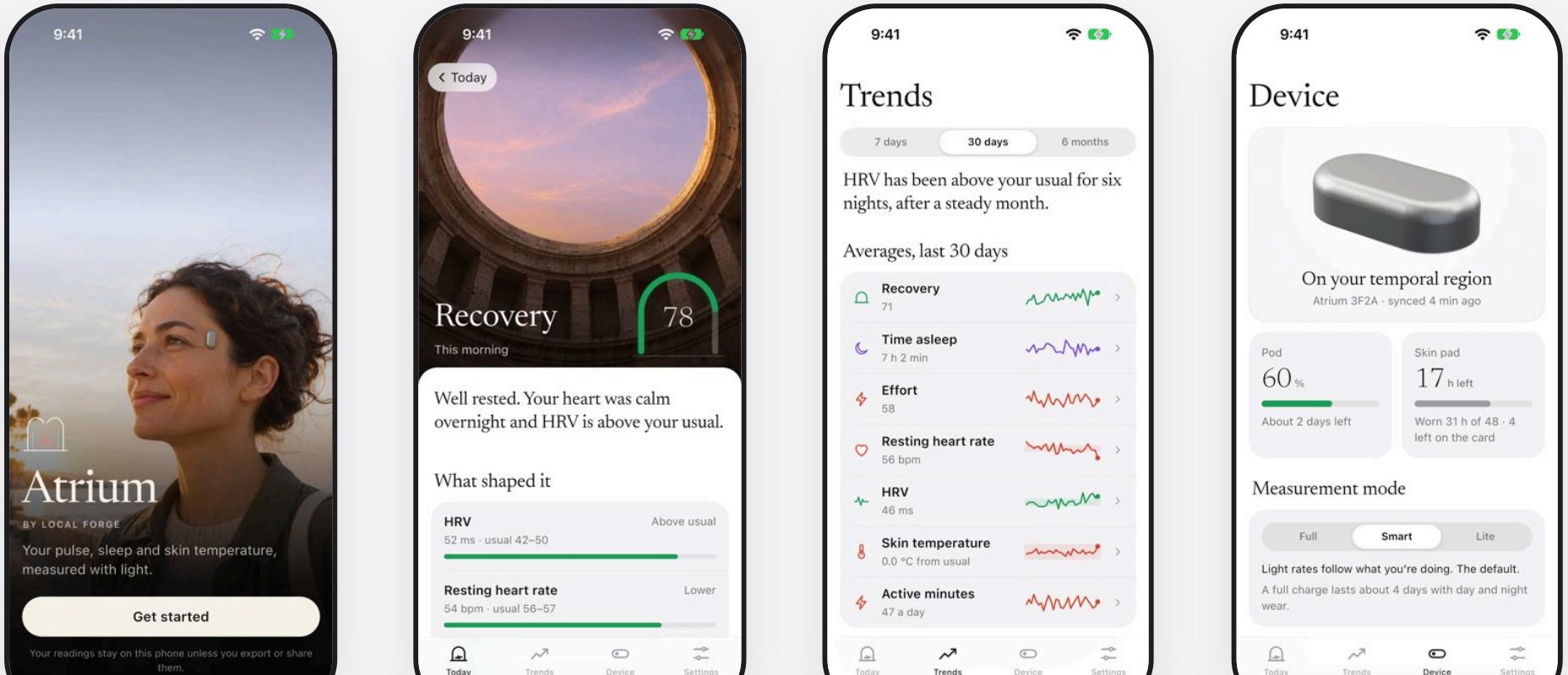
Your night, in one sentence.

A photo, one line, one number. Every score is measured against your own usual range.



Everything stays on your phone.

No sign-up. The pod pairs over encrypted Bluetooth and sends each reading every 30 seconds while it's near.



Intelligence

The pod measures. The phone understands the night.

On the pod

Heart rate, every second

Beat-to-beat HRV, every 30 s

Wear, motion and steps

Fit quality

Head angle and skin temperature

Oxygen ratio, in Labs

On the phone

Sleep and wake, from a model you can update

Resting heart rate and nightly HRV

Skin temperature against your 14-night baseline

Sleep position, and time spent on the pod

Steps and active minutes

Fit and wear-side checks

Wellness estimates, not a medical device. The sleep model was trained on wrist data and is retrained on pod nights before launch.

By the numbers.

189 µA
Average draw

5.3 days
Typical wear per charge

96 min
To 80 % on the Plinth

+22.8 dB
Median Bluetooth margin at 1 m

30 s
Per record, sent live

0 bytes
Sent to a cloud

Simulated on the EVT design; to be measured on EVT boards.

Cost at 1,000 pods

Priced so far.

$20.58

Per pod, list prices, no discount, ₹95 to the dollar.
$20,582 for 1,000 pods, about ₹19,55,249.

	Per pod	1,000 pods
Components, 52 parts	$18.43	$18,429
Cell, ₹150	$1.58	$1,579
Islands, 6-layer HDI, full (est.)	$0.44	$444
Flex jumper	$0.07	$72
Board shipping	$0.06	$58
With HDI boards	**$20.58**	**$20,582**
Through-via boards, after a re-route	$20.35	$20,349

Still to quote

Assembly
SMT on both islands and the flex, stencils, setup, extended-part fees

Shell
Moulded polycarbonate, NCVM finish, hardcoat

Plinth and skin pads
Moulded parts and die-cut adhesive

Charging case
Board and key parts priced, $1,001 for 1,000 with shipping; enclosure, 500 mAh cell and assembly to quote

Test, programming and yield
Plus packaging, freight, duty and GST

HDI estimate: JLC's instant quote is $322.78; it adds buried-via and lamination fees after a file review. Taken here as +25–50 %, $403–$484, mid-point used.

Where we are

Designed. Routed. Ready to build.

The capsule, both boards, the charging case, the firmware and the app are done in their first full form. What comes next turns them into a product you can wear.

- ● **Design locked**
  Capsule, finish and size approved
- ● **Boards routed**
  Pod and case, design rules clean
- ● **Firmware and app**
  Live sync, on-phone algorithms
- ○ **Close the board items**
  Optics model, flex bend, rib clearance
- ○ **Order the optics by their new code**
  MAXM86161AEFD+T, active at DigiKey
- ○ **First pods on the bench**
  Then the first real night

Open source

Every file. Open to everyone.

Schematics, board layouts, CAD, the bill of materials, firmware, the app and these documents. Build one, repair one, improve one.

CERN-OHL-S-2.0

Hardware

Schematics, layout, CAD and BOM. Strongly reciprocal: changes you ship stay open.

Apache-2.0

Software

Firmware, the app and the optional cloud. Use it anywhere, with a patent grant.

CC BY-SA 4.0

Documents

Docs and images. Share and adapt with credit, under the same terms.

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Appendix A

Pod bill of materials

Every line, 1,000 units, list prices, no discount. JLCPCB stock and price tiers via pcbparts, 8 November 2026; U3 from DigiKey; cell at ₹150 at ₹95 to the dollar; boards are whole orders spread over 1,000 pods; board shipping in the totals.

Qty	Ref	Part	Source	Stock	Unit	Per pod
1	U3	MAXM86161AEFD+T	DigiKey	3,037	$9.7308	**$9.731**
1	U1	NRF52832-CIAA-R	C509405	18,423	$3.6062	**$3.606**
1	U11	LSM6DSOXTR	C481766	9,495	$2.5465	**$2.546**
1	U10	BQ21061YFPR	C2869893	2,744	$0.8603	**$0.860**
2	U2, U13	TPS7A0218PYCHR	C3752366	118	$0.3451	**$0.690**
1	Y1	NX1612SA-32MHZ-STD-CIS-3	C910049	8,616	$0.2372	**$0.237**
1	C51	GRM155R60J226ME11D	C415703	123,060	$0.1252	**$0.129**
1	ANT1	KH-1608-H08	C2925416	2,983	$0.1224	**$0.122**
9	C9, C20, C21, C31, C32, C50, C52, C56, C63	GRM033R60J105KEA2D	C76929	1,489,217	$0.0091	**$0.084**
2	R10, R11	RT0201BRD0710KL	C852211	7,445	$0.0385	**$0.079**
3	D10, D12, D13	ESD5V3U04-0201	C521889	9,995	$0.0251	**$0.078**
1	L1	LQM18FN100M00D	C86083	259,865	$0.0550	**$0.057**
1	C8	GRM035R60J475ME15D	C335103	269,124	$0.0426	**$0.044**
3	TH1, TH5, TH4	NCP03XH103F05RL	C98098	9,441	$0.0119	**$0.037**
1	U12	AP6683	C2849565	101,423	$0.0257	**$0.026**
1	C30	CL05A106MQ5NUNC	C15525	9,182,955	$0.0192	**$0.020**
1	C3	GJM0335C1H1R0BB01D	C237514	39,117	$0.0159	**$0.016**
2	C1, C2	GRM0335C1H120GA01D	C161404	110,721	$0.0077	**$0.016**
5	C4, C5, C12, C60, C55	GRM033R60J104KE19D	C76928	1,571,832	$0.0021	**$0.011**
1	L3	LQP03TN3N3B02D	C98045	282,123	$0.0104	**$0.011**
1	C53	CL05A475MP5NRNC	C23733	1,033,890	$0.0092	**$0.009**
1	L2	VHF100505HQ15NJT	C395049	283,935	$0.0042	**$0.004**
2	R1, R2	RC0201FR-074K7L	C138135	3,323,433	$0.0018	**$0.004**
2	R5, R41	RC0201FR-071ML	C295786	2,330,029	$0.0012	**$0.002**
2	C6, C7	0201CG101J500NT	C62550	1,578,572	$0.0009	**$0.002**
1	R40	RC0201FR-072M2L	C474576	17,500	$0.0018	**$0.002**
1	R51	RC0201FR-07330RL	C274872	1,215,785	$0.0017	**$0.002**
1	R3	0201WMF1003TEE	C270364	2,679,121	$0.0011	**$0.001**
1	R50	0201WMF1002TEE	C473048	8,722,734	$0.0010	**$0.001**
1	Z41	0201WMF0000TEE	C473473	2,918,711	$0.0009	**$0.001**
1	BT1	Sharvi HT19 LiPo, 3.7 V 30 mAh	Sharvi	–	₹150	**$1.579**
1	FPC1	Flex jumper, 2-layer	JLCPCB	–	–	**$0.072**
2	Radio, power	6-layer islands, HDI full, est. (through-via $0.211)	JLCPCB	–	–	**$0.444**
7	P1–P5, TP1–TP2	Contact and test pads, ENIG copper on the boards	–	–	–	**$0.000**
		Components				**$18.43**
		Per pod with HDI boards, incl. board shipping				**$20.58**
		Per pod, through-via boards, incl. board shipping				**$20.35**

Appendix B

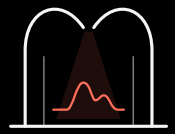
Boards and charging case

JLCPCB online quotes, 8 November 2026, list prices, excluding shipping (islands $31.69, flex $26.60, case $43.19).

Board	Build	Order	Price	Per unit
Radio + power islands, HDI	6-layer HDI 1-step, 0.8 mm, ENIG, 0.1 mm filled vias; instant quote	25 panels × 42 sets	$322.78	**$0.307**
Radio + power islands, HDI full	Plus buried-via and lamination fees, est. +25–50 %, mid-point	25 panels × 42 sets	$443.82	**$0.423**
Radio + power islands, no HDI	6-layer through-via, 0.8 mm, ENIG, 0.1 mm filled and capped	25 panels × 42 sets	$211.27	**$0.201**
Flex jumper	2-layer, 0.12 mm, ENIG, steel stiffener	20 panels × 55	$71.65	**$0.065**
Charging-case board	2-layer FR-4, 0.8 mm, ENIG, 0.2 mm vias	1,000 boards	$172.63	**$0.173**

HDI: JLC confirms buried-via and lamination fees only after a file review, so the HDI full line is an estimate until that quote. No HDI: 7–8 days, includes the 4-wire Kelvin test JLC requires for 0.1 mm vias. The current layout needs HDI: 13 of the nRF52832's 15 inner-ball vias would land on bottom-side pads, so a through-via version needs a re-route first. The case board's 0.2 mm vias add $76.83 per 1,000.

Case part	Role	Unit
CH32V003F4U6	MCU	**$0.1880**
TP4054	Charger	**$0.0375**
TPS61023	Boost	**$0.1732**
SY6280AAC	Switch	**$0.0527**
XC6206P182MR	1.8 V LDO	**$0.0982**
HAL248	Lid sensor	**$0.0861**
USB-C 16-pin	Receptacle	**$0.0509**
SKSCLAE010	Button	**$0.0982**
Key parts per case		**$0.78**



Atrium.

Open to the sky. Open to everyone.

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